

Fermat's last theorem for n=4

Def * Elliptic curve over $\mathbb{Q}$ is equation of form:
 $y^2 = ax^3 + bx^2 + cx + d$ ($a, b, c, d \in \mathbb{Q}$, $a \neq 0$)
 where RHS polynomial has no multiple root

Lem 1.5 For distinct $a, b, c \in \mathbb{Q}$ define B , $\tilde{C}$ and C
 $B = \{(u, v, w) \in \mathbb{Q} \times \mathbb{Q} \times \mathbb{Q} \mid u^2 + a = v^2 + b = w^2 + c\}$
 $\tilde{C} = \{(x, y) \in \mathbb{Q} \times \mathbb{Q} \mid y^2 = (x-a)(x-b)(x-c)\}$.
 $C = \{(x, y) \in \mathbb{Q} \times \mathbb{Q} \mid y^2 = (x-a)(x-b)(x-c), y \neq 0\}$.
 (1) $\exists$ mutually inverse maps $f: B \rightarrow C$, $g: C \rightarrow B$
 $f(u, v, w) = (u^2 + a + uv + vw + wu, (u+v)(v+w)(w+u),$
 $g(x, y) = \left(\frac{1}{2y}((x-a)^2 - (b-a)(c-a)), \right.$
 $\left. \frac{1}{2y}((x-b)^2 - (a-b)(c-b)), \frac{1}{2y}((x-c)^2 - (a-c)(b-c)) \right)$
 (2) $\exists$ map $h: B \rightarrow \tilde{C}$, $h(u, v, w) = (u^2 + a, uvw)$

Rem 1.6 For elliptic curve $y^2 = (x-a)(x-b)(x-c)$
 $h \circ g: C \rightarrow \tilde{C}$ is called multiplication-by-2 map

$a=d, b=0, c=-d, \begin{cases} f: B_d \rightarrow C_d \\ g: C_d \rightarrow B_d \end{cases}$

$i: (x, y, z) \mapsto \left(\frac{y-x}{2}, \frac{z}{2}, \frac{x+y}{2} \right)$
 $j: (u, v, w) \mapsto (w-u, w+u, 2v)$

$\begin{cases} i: A_d \rightarrow B_d \\ j: B_d \rightarrow A_d \end{cases}$

Lem 1.4 For $d \in \mathbb{Q}$ define:
 $A_d = \{(x, y, z) \in \mathbb{Q} \times \mathbb{Q} \times \mathbb{Q} \mid x^2 + y^2 = z^2, \frac{1}{2}xy = d\}$
 $B_d = \{(u, v, w) \in \mathbb{Q} \times \mathbb{Q} \times \mathbb{Q} \mid u^2 + d = v^2, v^2 + d = w^2\}$
 $C_d = \{(x, y) \in \mathbb{Q} \times \mathbb{Q} \mid y^2 = x^3 - d^2x, y \neq 0\}$.
 $\exists$ bijections between any 2 of A_d, B_d, C_d

$\begin{cases} (i) & \Rightarrow A_d \neq \emptyset \\ (ii) & \Rightarrow B_d \neq \emptyset \\ (iii) & \Rightarrow C_d \neq \emptyset \end{cases}$

Prop 0.13 The area of a right $\triangle$ whose sides are integers is not a $\square$

$\begin{matrix} (i) \\ \rightleftharpoons \\ d=1 \end{matrix}$

Lem 1.3 $d \in \mathbb{Q}$, these are equivalent:
 (i) $\exists$ rational number sides **right** $\triangle$ of area d
 (ii) $\exists$ 3 $\square$ s of rational numbers that form arithmetic progression of difference d
 (iii) $\exists$ rational solution to $y^2 = x^3 - d^2x$ other than $(x, y) = (0, 0)$ and $(\pm d, 0)$

$\begin{matrix} (iii) \\ \rightleftharpoons \\ d=1 \end{matrix}$

Prop 1.2 The only rational solutions to $y^2 = x^3 - x$ are $(x, y) = (0, 0)$ and $(\pm 1, 0)$

1.1 sol $\cdot z^2/y^6$, $\nexists$:
 $0 \neq \left(\frac{x^2 z}{y^3} \right)^2 = \left(\frac{z^2}{y^2} \right)^3 - \frac{z^2}{y^2}$

Prop 1.1 $\nexists$ solution (x, y, z) to $x^4 + y^4 = z^4$ satisfying $xyz \neq 0$